

# SAT Practice — Answer Key & Trap Analysis

Correct Answers, Explanations, and Your Personal Pitfall for Each Question

---

## Question 1 — Math

Advanced Math · Exponential Functions | Module 2 Hard

**Correct Answer:** C

**Explanation:** 100 minutes =  $\frac{5}{3}$  hours. Let  $x$  count the number of 100-minute periods, so  $t = (\frac{5}{3})x$ . Substituting into the exponent:  $(\frac{3}{5}) \cdot (\frac{5}{3})x = x$ . The function becomes  $N = 800 \cdot (1.05)^x$ , so each 100-minute period the population is multiplied by 1.05, which is a 5% increase. Therefore  $p = 5$ .

■ **YOUR LIKELY TRAP:** Picking D (= 60) or B (= 3) by pulling a number straight from the fractional exponent instead of substituting the correct time period — the same error pattern from PT4 Q10 where you chose 6.

---

## Question 2 — Math

Advanced Math · Quadratic Discriminant | Module 2 Hard

**Correct Answer:** 65

**Explanation:** Expand:  $2kx^2 - 96x + 18 = 0$ . So  $a = 2k$ ,  $b = -96$ ,  $c = 18$ . Discriminant =  $(-96)^2 - 4(2k)(18) = 9,216 - 144k$ . No real solutions requires  $9,216 - 144k < 0$ , giving  $k > 64$ . Since  $k$  must be a positive integer, the least value is 65.

■ **YOUR LIKELY TRAP:** Getting  $k > 64$  and writing 64 — the exact off-by-one boundary error from PT10 Q20 (writing 49 instead of 50) and PT4 Q13 (writing 289.1 instead of 289). When the discriminant equals zero the equation still has a real solution (a repeated root).

---

## Question 3 — Math

Problem-Solving & Data Analysis · Unit Conversion | Module 2 Hard

**Correct Answer:** B

**Explanation:** 1 m = 100 cm, so  $1 \text{ m}^2 = 10,000 \text{ cm}^2$ . Converting:  $(540 \text{ cm}^2/\text{sec}) \times (1 \text{ m}^2/10,000 \text{ cm}^2) \times (60 \text{ sec}/1 \text{ min}) = 540 \times 60 / 10,000 = 32,400/10,000 = 3.24 \text{ m}^2/\text{min}$ .

■ **YOUR LIKELY TRAP:** Dividing by 100 instead of 10,000 (forgetting to square the linear conversion factor), which gives 5.40 (choice C) or 324 (choice D). This mirrors PT9 Q21 where you divided by 3.28 instead of  $3.28^2$  for square units.

---

## Question 4 — Math

Advanced Math · Rational Expressions | Module 2 Hard

**Correct Answer:** A

**Explanation:** Factor the second denominator:  $x^2y - 5xy = xy(x - 5)$ . The second fraction simplifies:  $y(x-5) / [xy(x-5)] = 1/x$ . But we need common denominators with the first fraction. Multiply the first term by  $xy/xy$ :  $xy(y+9)/[xy(x-5)] + y(x-5)/[xy(x-5)]$ . Numerator:  $xy^2 + 9xy + xy - 5y = xy^2 + 10xy - 5y$ . Denominator:  $xy(x-5) = x^2y - 5xy$ . Answer: A.

■ **YOUR LIKELY TRAP:** Choosing B by multiplying the denominators  $(x-5) \times xy(x-5)$  instead of recognizing that  $xy(x-5)$  is already the LCD — the same failure to simplify/cancel before combining that led you to pick D on PT10 Q18.

---

## Question 5 — Reading & Writing

Craft and Structure · Words in Context | Module 2 Hard

**Correct Answer:** A

**Explanation:** 'Sanguine' means optimistic. The sentence contrasts economists who 'praised' the agreement with those who are less positive — less sanguine. The other choices describe qualities (careful, neutral, honest) that don't logically contrast with 'praised.'

■ **YOUR LIKELY TRAP:** Picking D ('forthright') because it sounds like a plausible intellectual quality, just as you picked 'earnest' over 'sanguine' on PT9 Q4. The key is identifying the **CONTRAST** structure: 'praised' vs. 'less \_\_\_\_ analysts.'

---

## Question 6 — Reading & Writing

Standard English Conventions · Verb Forms | Module 2 Hard

**Correct Answer:** A

**Explanation:** The sentence's main verb is 'has prompted.' The blank must be filled with a nonfinite verb form that creates a participial phrase modifying 'survey.' Only 'revealing' (a present participle) works here. The finite forms 'reveals,' 'revealed,' and 'has revealed' would each create a second main verb, making the sentence ungrammatical.

■ **YOUR LIKELY TRAP:** Choosing C ('revealed') or B ('reveals') because they sound complete and natural in isolation — the same confusion between finite and nonfinite verb forms that tripped you on PT9 Q19 (choosing 'suggests' instead of 'suggesting') and PT4 Q20 ('to be' instead of 'will be').

---

## Question 7 — Reading & Writing

Standard English Conventions · Punctuation | Module 2 Hard

**Correct Answer:** C

**Explanation:** A comma is needed to set off the supplementary noun phrase 'a discovery that...' from the main clause. A semicolon (A) cannot join a main clause to a noun phrase. A period (B) creates a sentence fragment ('A discovery that...'). Omitting punctuation (D) creates a run-on.

■ **YOUR LIKELY TRAP:** *Choosing A (semicolon) because the supplementary element feels long enough to be its own clause — the same error from PT7 Q18 where you chose the semicolon instead of the comma before a supplementary appositive.*

---

## Question 8 — Reading & Writing

Information and Ideas · Inference / Logical Completion | Module 2 Hard

**Correct Answer:** A

**Explanation:** The passage explains that the model prescribes uniform humidity because actual data is too sparse — a known simplification. When a model uses a simplification, some disagreement between its outputs and reality is expected. Choice A logically follows. B introduces an unrelated claim about future accuracy. C asserts a specific direction of error not supported by the text. D explains internal inconsistencies, not the model-vs-reality gap the passage sets up.

■ **YOUR LIKELY TRAP:** *Choosing C or D because they sound scientifically specific, just as you picked C instead of A on PT11 Q14 (the Titan methane mole fraction question). The passage sets up a known-limitation → expected-consequence pattern; the correct answer states that consequence directly rather than speculating about specifics the text doesn't support.*

---